

Multiple Choice (10 marks)

1. What is one condition necessary to produce diamonds?
 - (a) high altitude
 - (b) intense pressure
 - (c) quick cooling
 - (d) low temperatures
2. Heart muscle is best described as
 - (a) the same kind of cells working together.
 - (b) different systems working together.
 - (c) the same kind of organs forming a system.
 - (d) different organs performing a function.
3. Which of the following is NOT a description of compounds?
 - (a) They can exist in the form of atoms or molecules.
 - (b) They consist of atoms of two or more elements bonded together.
 - (c) They have properties that are different from their component elements.
 - (d) They can be broken down into elements by chemical means but not physical means.
4. In a copper wire, a temperature increase is the result of which of the following?
 - (a) an increase in the size of the copper particles
 - (b) a decrease in the mass of the copper particles
 - (c) an increase in the motion of the copper particles
 - (d) a decrease in the distance between the copper particles
5. The gravitational force of the Sun affects the planets in our solar system. Which of these is influenced the most by this force?
 - (a) axial tilt
 - (b) orbital path
 - (c) the masses of the planets
 - (d) the number of moons per planet

True / False (10 marks)

Answer True or False

6. True or False: Hot air warming a room is an example of heat transferred by radiation.
7. True or False: If a nerve cell stops functioning, it will stop sending signals to the brain.
8. True or False: A candle providing light primarily represents the use of mechanical energy through its flame and heat production.
9. True or False: In January, the constellation Gemini is visible at noon when the sun is highest in the sky.
10. True or False: The distance from the Sun to Earth is approximately 2.0 light-seconds.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the types of evidence that could challenge the claim of extinction for a bird species last seen in the 1900 s, and analyze how such evidence might impact conservation efforts.
12. Discuss the role of periodic forest fires in maintaining the stability of climax fire ecosystems and analyze the potential consequences of preventing these natural fires.

End of Paper